MSG No. 49.273 $P_{c}ccm$ [Type IV, orthorhombic]

Table 1: Wyckoff site: 2a, site symmetry: m'm'm

No.	position	mapping
1	$[0, 0, 0]$	[1,4,5,8,10,11,14,15]
2	$\left[0, 0, \frac{1}{2}\right]$	[2,3,6,7,9,12,13,16]

Table 2: Wyckoff site: 2b, site symmetry: m'm'm

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,5,8,10,11,14,15]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,3,6,7,9,12,13,16]

Table 3: Wyckoff site: 2c, site symmetry: m'm'm'

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,2,3,4,13,14,15,16]
2	$\left[0, 0, \frac{3}{4}\right]$	[5,6,7,8,9,10,11,12]

Table 4: Wyckoff site: 2d, site symmetry: m'm'm'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[1,2,3,4,13,14,15,16]
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[5,6,7,8,9,10,11,12]

Table 5: Wyckoff site: 2e, site symmetry: m'm'm

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,4,5,8,10,11,14,15]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,6,7,9,12,13,16]

Table 6: Wyckoff site: 2f, site symmetry: m'm'm

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,4,5,8,10,11,14,15]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,6,7,9,12,13,16]

Table 7: Wyckoff site: 2g, site symmetry: m'm'm'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[1,2,3,4,13,14,15,16]

2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,7,8,9,10,11,12]`

Table 8: Wyckoff site: `2h`, site symmetry: `m'm'm'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,7,8,9,10,11,12]`

Table 9: Wyckoff site: `4i`, site symmetry: `2'm'm`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,8,10,15]`
2	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[2,7,9,16]`
3	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[3,6,12,13]`
4	$[-x,\ 0,\ 0]$	`[4,5,11,14]`

Table 10: Wyckoff site: `4j`, site symmetry: `2m'm'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,2,15,16]`
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[3,4,13,14]`
3	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[5,6,11,12]`
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,10]`

Table 11: Wyckoff site: `4k`, site symmetry: `2'm'm`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,8,10,15]`
2	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,7,9,16]`
3	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,12,13]`
4	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[4,5,11,14]`

Table 12: Wyckoff site: `4l`, site symmetry: `2m'm'`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,15,16]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,4,13,14]`
3	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,11,12]`
4	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,8,9,10]`

Table 13: Wyckoff site: `4m`, site symmetry: `m'2'm`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,8,11,14]`
2	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[2,7,12,13]`
3	$\left[0,\ y,\ \frac{1}{2}\right]$	`[3,6,9,16]`
4	$[0,\ -y,\ 0]$	`[4,5,10,15]`

Table 14: Wyckoff site: `4n`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,3,14,16]`
2	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[2,4,13,15]`
3	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[5,7,10,12]`
4	$\left[0,\ y,\ \frac{3}{4}\right]$	`[6,8,9,11]`

Table 15: Wyckoff site: `4o`, site symmetry: `m'2'm`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ 0\right]$	`[1,8,11,14]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[2,7,12,13]`
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[3,6,9,16]`
4	$\left[\frac{1}{2},\ -y,\ 0\right]$	`[4,5,10,15]`

Table 16: Wyckoff site: `4p`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[1,3,14,16]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	`[2,4,13,15]`
3	$\left[\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	`[5,7,10,12]`
4	$\left[\frac{1}{2},\ y,\ \frac{3}{4}\right]$	`[6,8,9,11]`

Table 17: Wyckoff site: `4q`, site symmetry: `m'm'2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,14,15]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[2,3,13,16]`
3	$[0,\ 0,\ -z]$	`[5,8,10,11]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[6,7,9,12]`

Table 18: Wyckoff site: `4r`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,14,15]`

continued ...

Table 18

No.	position	mapping
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3,13,16]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5,8,10,11]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7,9,12]`

Table 19: Wyckoff site: `4s`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4,14,15]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[2,3,13,16]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,8,10,11]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[6,7,9,12]`

Table 20: Wyckoff site: `4t`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4,14,15]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3,13,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5,8,10,11]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7,9,12]`

Table 21: Wyckoff site: `8u`, site symmetry: `m'..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,14]`
2	$\left[0,\ -y,\ \frac{1}{2}-z\right]$	`[2,13]`
3	$\left[0,\ y,\ \frac{1}{2}-z\right]$	`[3,16]`
4	$[0,\ -y,\ z]$	`[4,15]`
5	$[0,\ -y,\ -z]$	`[5,10]`
6	$\left[0,\ y,\ z+\frac{1}{2}\right]$	`[6,9]`
7	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[7,12]`
8	$[0,\ y,\ -z]$	`[8,11]`

Table 22: Wyckoff site: `8v`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ z\right]$	`[1,14]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	`[2,13]`
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[3,16]`
4	$\left[\frac{1}{2},\ -y,\ z\right]$	`[4,15]`
5	$\left[\frac{1}{2},\ -y,\ -z\right]$	`[5,10]`
6	$\left[\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	`[6,9]`
7	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[7,12]`

continued ...

Table 22

No.	position	mapping
8	$\left[\frac{1}{2}, y, -z\right]$	[8,11]

Table 23: Wyckoff site: 8w, site symmetry: .m'.

No.	position	mapping
1	$[x, 0, z]$	[1,15]
2	$\left[x, 0, \frac{1}{2} - z\right]$	[2,16]
3	$\left[-x, 0, \frac{1}{2} - z\right]$	[3,13]
4	$[-x, 0, z]$	[4,14]
5	$[-x, 0, -z]$	[5,11]
6	$\left[-x, 0, z + \frac{1}{2}\right]$	[6,12]
7	$\left[x, 0, z + \frac{1}{2}\right]$	[7,9]
8	$[x, 0, -z]$	[8,10]

Table 24: Wyckoff site: 8x, site symmetry: .m'.

No.	position	mapping
1	$\left[x, \frac{1}{2}, z\right]$	[1,15]
2	$\left[x, \frac{1}{2}, \frac{1}{2} - z\right]$	[2,16]
3	$\left[-x, \frac{1}{2}, \frac{1}{2} - z\right]$	[3,13]
4	$\left[-x, \frac{1}{2}, z\right]$	[4,14]
5	$\left[-x, \frac{1}{2}, -z\right]$	[5,11]
6	$\left[-x, \frac{1}{2}, z + \frac{1}{2}\right]$	[6,12]
7	$\left[x, \frac{1}{2}, z + \frac{1}{2}\right]$	[7,9]
8	$\left[x, \frac{1}{2}, -z\right]$	[8,10]

Table 25: Wyckoff site: 8y, site symmetry: ..m

No.	position	mapping
1	$[x, y, 0]$	[1,8]
2	$\left[x, -y, \frac{1}{2}\right]$	[2,7]
3	$\left[-x, y, \frac{1}{2}\right]$	[3,6]
4	$[-x, -y, 0]$	[4,5]
5	$\left[x, y, \frac{1}{2}\right]$	[9,16]
6	$[x, -y, 0]$	[10,15]
7	$[-x, y, 0]$	[11,14]
8	$\left[-x, -y, \frac{1}{2}\right]$	[12,13]

Table 26: Wyckoff site: 8z, site symmetry: ..m'

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	[1,16]

continued ...

Table 26

No.	position	mapping
2	$\left[x,\, -y,\, \frac{1}{4}\right]$	[2,15]
3	$\left[-x,\, y,\, \frac{1}{4}\right]$	[3,14]
4	$\left[-x,\, -y,\, \frac{1}{4}\right]$	[4,13]
5	$\left[-x,\, -y,\, \frac{3}{4}\right]$	[5,12]
6	$\left[-x,\, y,\, \frac{3}{4}\right]$	[6,11]
7	$\left[x,\, -y,\, \frac{3}{4}\right]$	[7,10]
8	$\left[x,\, y,\, \frac{3}{4}\right]$	[8,9]

Table 27: Wyckoff site: 16A, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[2]
3	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[6]
7	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[7]
8	$[x,\, y,\, -z]$	[8]
9	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[9]
10	$[x,\, -y,\, -z]$	[10]
11	$[-x,\, y,\, -z]$	[11]
12	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[13]
14	$[-x,\, y,\, z]$	[14]
15	$[x,\, -y,\, z]$	[15]
16	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[16]